Physical Account -- 8-Class Resource Ledger & Errata Verification (1 Page)

1. Subcomponent Mass, Position & Stability Ledger (Organizer Errata Applied)

Stability is computed strictly per subcomponent and assigned position (x1.0 main deck bays bay1/bay2, x1.2 hangar/hangar_roof/flight_deck, x3.0 mast_3), never by multiplying an entire multi-part guest by the mast factor. Local structural limits (Bay 1 <= 8,000 kg, Bay 2 <= 4,000 kg, Hangar <= 1,000 kg, VTOL airframe <= 150 kg, mast_3 <= 60 kg / 0.6 m2, hangar_roof <= 150 kg / 2.0 m2) are enforced alongside global limits (10,000 kg mass, 12,000 kg-eq stability). Under the organizer errata, G4 hangar station is 850 kg (overriding PDF 900 kg) + 140 kg drone = 990 kg, and G5 includes both the 3,000 kg container and 25 kg antenna = 3,025 kg:

Guest ID & Official Name	Subcomponents & Assigned Positions	Mass (kg)	Stability Multiplier & Calculation	Stability (kg-eq)	Electrical, Thermal & Data Envelope
G1 -- Passive Radio & Meteo Sensor	G1_MAST_SENSOR_HEAD -> mast_3 (0.5 m2)	45	45 x 3.0 (mast_3)	135.0	1.5 kW air on mast_3 (<= 2.0 kW limit; 0 W water); 50 Mbps LAN, 200 kbps SATCOM
G2 -- Ocean & Weather Module	G2_ROV_WINCH_CONTAINER_10FT -> bay2 / bay1_a	3,500	3,500 x 1.0 (deck bay)	3,500.0	20 kW sustained 300 s winch / 6 kW continuous -> water; 5 Mbps LAN, 500 kbps SATCOM
G3 -- Seabed Inspection AUV + Cradle	G3_AUV_CRADLE_BAY1_20FT -> bay1 (20 ft via ramp)	5,500	5,500 x 1.0 (deck bay)	5,500.0	1 kW survey / 25 kW (3,600 s) uninterruptible charge -> water; 100 Mbps cradle LAN
G4 -- VTOL Rescue Drone + Hangar (Errata)	G4_DRONE_VEHICLE (140 kg) + G4_HANGAR_STATION (850 kg) -> hangar	990	(140 + 850) x 1.2 (hangar <= 1,000 kg)	1,188.0	35 kW (1,200 s) recovery/charge, 10 kW patrol, 2 kW standby (0 bps silence); 20 Mbps LAN
G5 -- Emergency Comms & Edge Compute	G5_COMPUTE_CONTAINER (3,000 kg -> bay2/bay1_a) + G5_MAST_ANTENNA (25 kg -> mast_3/hangar_roof)	3,025	3,000 x 1.0 + 25 x 3.0 (mast_3) or + 25 x 1.2 (roof)	3,075.0 (3,030.0 roof)	45 kW (43.8 kW water + 1.2 kW air) / 8 kW degraded (30% retention); 1 Gbps LAN, 6 Mbps SATCOM
G6 -- Civil Hydrographic & eDNA Micro-Sampler	G6_HYDRO_SKID_10FT -> bay2 / bay1_a (All fields ASSUMED, port_only=true)	420	420 x 1.0 (deck bay)	420.0	4.5 kW (1,800 s) / 1.5 kW eco -> water; 30 Mbps LAN, 200 kbps SATCOM

2. Five Combinations of Four Official Guests ({G1..G5}): Mass vs. Full 8-Class Admission

Evaluating all C(5,4) = 5 four-guest combinations confirms the organizer errata (exact 3 of 5 exceed the 10,000 kg margin) and proves why passing mass does not imply passing placement, mast/roof clearance, or 14-day Mission Envelope A.10 constraints:

4-Guest Set	Total Mass (kg)	> 10,000 kg?	Total Stability (kg-eq)	Physical Placement, Mast/Roof & Mission Envelope Verdict (D1)
{G1, G2, G3, G4}	10,035 kg	YES (+35 kg)	10,323.0 (<= 12,000)	YES_IF (Exceeds 10,000 kg by 35 kg; displacing G2 yields 6,535 kg)
{G1, G2, G3, G5}	12,070 kg	YES (+2,070 kg)	12,210.0 (> 12,000)	YES_IF (Exceeds mass & stability; G1 & G5 collide on mast_3 and 1.5+1.2=2.7 kW > 2.0 kW mast air)
{G1, G2, G4, G5}	7,560 kg	NO (2,440 kg free)	7,898.0 (<= 12,000)	YES_IF (Passes mass/stability, but G5 antenna collides with G1 on mast_3 or requires G4 hangar_roof clearance)
{G1, G3, G4, G5}	9,560 kg	NO (440 kg free)	9,898.0 (<= 12,000)	YES_IF (Passes mass/stability with G5 container in bay2, but G5 antenna collides on mast_3 / hangar_roof)
{G2, G3, G4, G5}	13,015 kg	YES (+3,015 kg)	13,263.0 (> 12,000)	YES_IF (Exceeds mass & stability; G2+G5 containers cannot share bay2 (6,500 kg > 4,000 kg) while G3 occupies bay1)

(Note: Replacing G5 with custom card G6 in {G1, G2, G4, G6} yields 4,955 kg, 5,243.0 kg-eq, G1 on mast_3, G2 on bay2, G4 in hangar, and G6 in bay1_a -> YES across all 8 resource classes.)

3. Cooling Curve, Thermal Inertia & Scenario B Power Accounting

* Chilled-Water Curve: C(T) = 160,000 - 40,000 * (T - 12) / 18 W for T in [12, 30] deg C. At 30 deg C, C(30) = 120 kW nominal; with -40% loop degradation, C_deg(30) = 72 kW. Sustained loads (> 60 s) cannot borrow from the +30 kW / 60 s (1.8 MJ) thermal inertia buffer.

* Scenario B Power vs. Cooling Binding: When D3 approves G4 recovery at T=480 s, the governor reserves G4's committed 35 kW recovery/charging load alongside G3's firm 25 kW post-leg recharge reservation. When generator reserve drops to 100 kW at T=720 s, instantaneous draw is 81 kW <= 100 kW during G4 recovery, whereas firm committed power is 25 + 35 + 45 = 105 kW > 100 kW (5 kW firm deficit), and at 30 deg C degraded loop (72 kW), chilled-water load is 1 + 35 + 43.8 = 79.8 kW > 72 kW (7.8 kW thermal deficit). Stepping down G5 (45 kW -> 8 kW) at T=720 s saves 37 kW power and 36.3 kW water cooling, maintaining 68 kW <= 100 kW (32 kW free margin) and zero infeasible ticks across the entire timeline.

4. What would change if the vessel had one single platform modification (mast_4)?

If a single structural modification were made to the vessel -- adding a second guest mast slot mast_4 (<= 60 kg, <= 0.6 m2, <= 2.0 kW independent forced-air cooling, x3.0 stability multiplier) -- the structural placement and mast air-cooling bottleneck blocking both mass-feasible four-guest combinations {G1, G2, G4, G5} (7,560 kg, 7,898.0 kg-eq) and {G1, G3, G4, G5} (9,560 kg, 9,898.0 kg-eq) is eliminated: G1_MAST_SENSOR_HEAD (45 kg, 1.5 kW air on mast_3) and G5_MAST_ANTENNA (25 kg, 1.2 kW air on mast_4) occupy separate mast slots without exceeding the 2.0 kW per-slot air-cooling limit or encroaching on G4's hangar_roof VTOL rotor clearance. Both combinations then achieve feasible static placement (YES_IF under Mission Envelope A.10), subject only to operational mode scheduling: (a) during degraded SATCOM (2.0 Mbps) or EMCON silence (W4, 0 bps), G5 (6.0 Mbps nominal SATCOM) steps down to DEGRADED_70PCT_LOSS / offline buffer (can_buffer_offline=true), and (b) during W2/W3/W5, G2/G3/G4 deck, acoustic-quiet (forbids_deck_machinery_while_active), and speed/heading windows remain time-multiplexed by the D3/D4 governor.